ALLEN

(Academic Session: 2026 - 2027)

TALLENTEX PRACTICE QUESTIONS

CLASS 10

CHEMISTRY

Your Hard Work Leads to Strong Foundation

Practice Questions

1. When two liquids are mixed, a solution with a pH value of 7 is formed. What are the pH values of the two liquids?

(1) 5, 2 (2) 3, 12 (3) 6, 1 (4) 14, 7

2. The comparison of the particles P, Q, R, S and T is given in the table

Substance	No. of Protons	No. of Neutrons	No. of electrons
P	25	30	25
Q	13	13	13
R	13	14	13
S	9	10	9
T	9	10	10

Identify atoms, ions and isotopes.

	Atoms	Ion	Isotopes
(1)	P, Q, R	R, T	S and T
(2)	R, T	P, Q, S	Q and R
(3)	P, Q, R, S	T	Q and R
(4)	Q, R	P, T	S and T

3. Assertion (A): Lead nitrate on thermal decomposition gives lead oxide, brown coloured nitrogen dioxide and oxygen gas.

Reason (R): Lead nitrate reacts with potassium iodide to form white ppt of lead iodide.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true and R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

4. Plaster of Paris $\left(CaSO_4.\frac{1}{2}H_2O\right)$ on mixing with water sets to form

(1) $CaSO_4.H_2O$ (2) $CaSO_4.1\frac{1}{2}H_2O$ (3) $CaSO_4.2H_2O$ (4) $CaSO_4.2\frac{1}{2}H_2O$

5. The valency of which one of the following is different from the others.

(1) Potassium ion (2) Barium ion

(3) Lithium ion (4) Ammonium ion

6. The metal obtained by self-reduction process is

(1) Cu (2) Zn (3) Fe (4) Ag

7. Read the given statements and mark the correct option.

Statement 1 : Bohr's orbits are called stationary orbits.

Statement 2 : Electrons remain stationary in these orbits for some time.

(1) Both statements 1 and 2 are true and statement 2 is the correct explanation of statement 1.

(2) Both statements 1 and 2 are true but statement 2 is not the correct explanation of statement 1.

(3) Statement 1 is true and statement 2 is false.

(4) Both statements 1 and 2 are false.

8. Which of the following are isobars?

(a) $_1H^1$, $_1H^2$, $_1H^3$ (b) $_{20}Ca^{40}$, $_{18}Ar^{40}$ (c) $_6C^{12}$, $_6C^{13}$, $_6C^{14}$ (d) $_1H^3$, $_2He^3$

(1) (a) and (c) (2) (a) and (d) (3) (b) and (d) (4) (b) and (c)

9. When 42 g of white solid 'X' is heated, 11 g of a gas 'P' and 5 g of gas 'Q' are evolved leaving behind a solid residue 'R' of mass 26 g.

'P' is an acidic gas which turns lime water milky. 'Q' is a neutral gas which condenses into a liquid and turns anhydrous $CuSO_4$ blue in colour. The aqueous solution of 'R' is alkaline to litmus and gives white precipitate 'S' with $BaCl_2$ solution.

Compounds 'X' and 'R' respectively are

(1) Na_2CO_3 and $NaHCO_3$ (2) $NaHCO_3$ and Na_2CO_3

(3) K_2S and K_2SO_3 (4) $KHCO_3$ and K_2CO_3

10. An ion of an atom has a net charge +3. It has 10 electrons and 14 neutrons, find the mass number of atom.

(1) 24 (2) 23 (3) 27 (4) 26

11. The schematic atomic structures of three elements X, Y and Z are given as



Which of the following statements is/are incorrect?

(I) Z can form ZCl_3 and ZCl_5.

(II) Y can exist in triatomic form.

(III) X and Y combine to form X_3Y.

(IV) X and Z combine to form X_3Z.

(V) X will gain one electron to form a stable compound.

(VI) X and Y combine to form X_2Y_2, Z and Y combine to form Y_3Z_2.

(1) I, II, IV, V (2) I, II, III, IV, V (3) I, III, IV, V, VI (4) I, II, V, VI

12. Which of the following is the formula of barium peroxide and barium oxide respectively?

(1) Ba_2O, Ba_2O_2 (2) Ba_2O_2, Ba_2O (3) BaO_2, BaO (4) BaO_3, BaO

13. An ionic bond is a type of chemical bond in which two oppositely charged ions are held through electrostatic force. The atomic numbers of four elements K, L, M and N are 6, 8, 10 and 12 respectively. Which two elements out of these can react to form ionic compound?

(1) K and M (2) L and N (3) K and N (4) L and M

14. The correction increasing order of molecular weights is.

(1) $H_2O > H_2S > CO_2 > SO_2$ (2) $H_2O < CO_2 < H_2S < SO_2$

(3) $H_2O < H_2S < CO_2 < SO_2$ (4) $H_2O < H_2S < SO_2 < CO_2$

15. A brief information about two atoms X and Y are given:

X: Atomic number = 7, Mass number = 14

Y: Atomic number = 7, Mass number = 15

Which of the following is correct about these two atoms?

(1) Electronic configuration of X is 2, 8, 4 while that of Y is 2, 8, 5.

(2) Both X and Y contain 7 neutrons.

(3) X has 2 electron shells while Y has 3 electron shells.

(4) Both X and Y have 5 valence electrons.

16. Neutron was discovered by

(1) Rutherford (2) Bohr (3) Chadwick (4) Goldstein

17. Which one of the following metal oxides shows both acidic and basic character ?

(1) CuO (2) K_2O (3) Na_2O (4) Al_2O_3

18. Which of the following statements is INCORRECT in accordance with a balanced chemical equation?

(l) It represents the exact number of atoms and molecules taking part in a chemical reaction.

(2) It does not tell anything about the speed of reaction.

(3) It reveals if any reaction is explosive.

(4) None of these.

19. Which of the following reaction is not a precipitation reaction?

(1) $AgNO_3 + NaCl \longrightarrow AgCl + NaNO_3$

(2) $Pb(NO_3)_2 + 2KI \longrightarrow PbI_2 + 2KNO_3$

(3) $FeCl_3 + 3NH_4OH \longrightarrow Fe(OH)_3 + 3NH_4Cl$

(4) $2KI + Cl_2 \longrightarrow 2KCl + I_2$

20. One molecule of an "ic" acid of a non-metal having 5 electrons in valence shell reacts with a molecule of base to form a salt "X." The base corresponds to the metal with one electron in valence shell. If the salt so formed can react with the same base in 1 : 2 ratio, predict the formula of the salt "X".

(1) K_2SO_4 (2) Na_2HPO_4 (3) KH_2PO_4 (4) Na_3PO_4

21. Take about 1g CaCO3 in a test tube heat over a flame when a colourless gas comes out of reaction.

Which type of reaction is this?

(1) Decomposition reaction (2) Displacement reaction

(3) Double displacement reaction (4) Combination reaction

22. An ionic compound of element 'X' and chlorine has the formula XCl2 having the molecular mass 95u.

What is the molecular mass of oxide of the element 'X' ?

(1) 56u (2) 40u (3) 64u (4) 80u

23. Which of the following statements about the electron is incorrect?

(1) It is a negatively charged particle.

(2) The mass of electron is equal to the mass of neutron.

(3) It is a basic constituent of all atoms.

(4) It is a constituent of cathode rays.

24. The chemical formulae of some substances are listed -

(a) AlF_3 (b) CO (c) CoO

(d) $NaHCO_3$ (e) Na_2S (f) N_2O_5

(g) $CaSO_4$ (h) $CuSO_3$

Which of the following has polyatomic anion with name ending in 'ate' ?

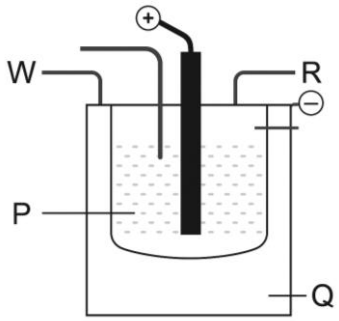
(1) (a), (b), (c) & (h) only (2) (a), (e), (f), (g) & (h) only

(3) (d) & (g) only (4) All (a) to (h)

25. Identify the incorrect match from the following.

		Turmeric	**Phenolphthalein**	**Methyl orange**
(1)	Lime water	Reddish brown	Colourless	Pink
(2)	Sugar solution	No change	No change	No change
(3)	Vinegar	Yellow	Colourless	Red
(4)	Baking soda	Reddish brown	Pink	Yellow

26. Below diagram is the arrangement of electrolytic cell.



In the above electrolytic method, 'P' undergoes electrolysis and gives 'Q' and gases 'R' and 'W'. 'W' when reacts with Nitrogen gas gives 'S'. 'S' when reacts with 'P' along with CO_2, forms 'T' and 'U' in Solvay process.

'R' along with slaked lime is further used in the manufacturing of 'V' in Hasenclever's plant. Identify P, Q, R, S, T, U and V.

	P	**Q**	**R**	**S**	**T**	**U**	**V**
(1)	$NaHCO_3$	CO_2	NH_3	NaCl	Na_2CO_3	$NaHCO_3$	$Ca(OH)_2$
(2)	NaCl(aq.)	NaOH	Cl_2	HNO_3	NH_4Cl	Na_2CO_3	$CaOCl_2$
(3)	NaCl (aq.)	NaOH	Cl_2	NH_3	NH_4Cl	$NaHCO_3$	$CaOCl_2$
(4)	NaOH	Cl_2	NaCl	NH_3	$NaHCO_3$	NH_4Cl	$Ca(OH)_2$

27. Three elements X, Y and Z contains 8, 16 and 17 protons respectively. The number of electrons when these elements will form their most stable anion will be respectively

(1) 8, 16, 17 (2) 10, 18, 18 (3) 8, 18, 18 (4) 10, 16, 16

28. Which of the following statements is always correct in case of neutral atom?

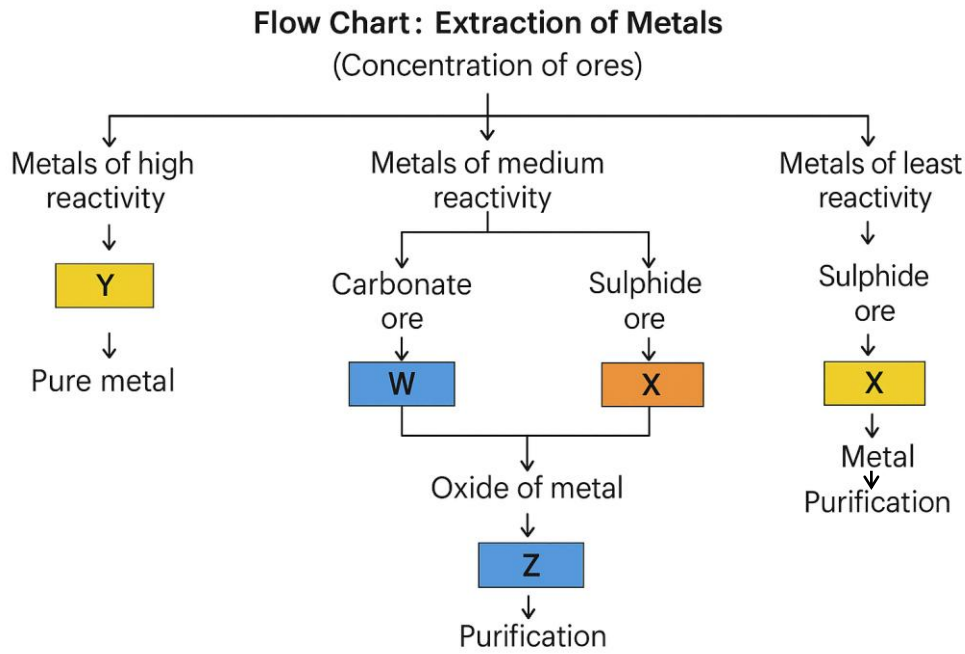
(1) An atom has equal number of electrons and protons.

(2) An atom has equal number of protons and neutrons.

(3) An atom has equal number of electrons and neutrons.

(4) An atom has equal number of electrons, protons and neutrons.

29. Namah, a student of class - 10th has prepared a flow chart for extraction of metals from their ores, but has left some gaps due to some confusion, which have been labelled as W, X, Y and Z.



Identify W, X, Y and Z in the flow chart.

	W	X	Y	Z
(1)	Roasting	Calcination	Electrolysis of molten ore	Reduction to metal
(2)	Roasting	Calcination	Electrolysis of molten ore	Oxidation of ore
(3)	Calcination	Roasting	Electrolysis of molten ore	Reduction to metal
(4)	Calcination	Cl_2	NaCl	Oxidation of ore

30. Arrange the following metals in the order of their decreasing reactivity.

K, Fe, Cu, Zn, Ca, Mg, Ag

(1) K > Ca > Zn > Mg > Cu > Ag > Fe

(2) K > Zn > Mg > Cu > Ca > Fe > Ag

(3) K > Mg > Ca > Zn > Fe > Cu > Ag

(4) K > Ca > Mg > Zn > Fe > Cu > Ag

31. There are four mixtures (i), (ii), (iii) and (iv) containing

i : Ammonium sulphate + Water

ii : Ammonium nitrate + Water

iii : Anhydrous Copper sulphate + Water iv : Glucose + Water

Identify the correct option.

(1) i - endothermic, ii - endothermic, iii - exothermic, iv - endothermic

(2) i - endothermic, ii - endothermic, iii - exothermic, iv - exothermic

(3) i - exothermic, ii - endothermic, iii - endothermic, iv - exothermic

(4) i - endothermic, ii - exothermic, iii - exothermic, iv – endothermic

32. Some elements and ions with their valencies are enlisted in the given table:

	Name (Valency)		**Name (Valency)**
(i)	Oxygen (2)	(vi)	Sodium (2)
(ii)	Iodine (1)	(vii)	Helium (2)
(iii)	Argon (0)	(viii)	Chlorine (1)
(iv)	Magnesium (3)	(ix)	Potassium (1)
(v)	Carbon (4)	(x)	Aluminium (3)

Select the incorrect match.

(1) (iv), (v), (vii) and (ix) (2) (iv), (vi) and (vii)

(3) (i), (viii), (ix) and (x) (4) (ii), (iii), (v) and (ix)

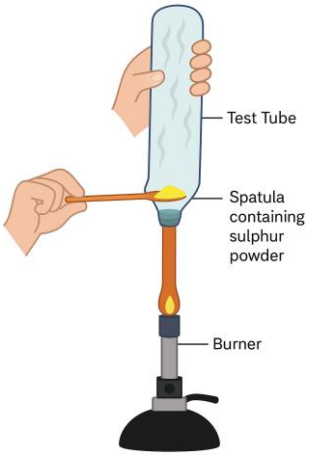
33. Which among the following pairs are having same number of electrons?

(1) O^{2-}, F^{-} (2) Mg^{2+}, Ar (3) P^{5+}, S^{2-} (4) Na, Al^{3+}

34. Carbon dioxide, made up of elements carbon and oxygen, contains 27.2% by mass of carbon. What mass of oxygen will combine with 10g of carbon?

(1) 26.7g (2) 3.74g (3) 7.28g (4) 4.28g

35. Pratyush took sulphur powder on a spatula and heated it. He collected the gas evolved by inverting a test tube over it as shown in fig. below. What will be the action of gas on (i) dry blue litmus paper ? (ii) moist blue litmus paper ?



(1) Dry blue litmus paper would not show any change and moist blue litmus paper would not show any change.

(2) Dry blue litmus paper would turn into red and moist blue litmus paper would turn into yellow.

(3) Dry blue litmus paper would not show any change and moist blue litmus paper will change into red.

(4) Dry blue litmus paper would not show any change and moist blue litmus paper will change into green.

36. In the Solvay process, when CO_2 is passed into brine with ammonia, then a product 'X' is produced. Thermal decomposition of this product 'X' will give 'Y' along with a gas and H_2O. Then the product 'Y' is -

(1) Na_2CO_3 (2) $NaHCO_3$ (3) NH_4Cl (4) NaOH

37. Approximate pH values of a few solutions are given :

Solution I : 2.2 – 2.4

Solution II : 6.8

Solution III : 7.4

Solution IV : 10

Solutions I, II, III and IV could be respectively

(1) Vinegar, Hydrochloric acid, Window cleaner and Milk

(2) Curd, Sodium hydroxide, Milk of magnesia and Coffee

(3) Lemon juice, Milk, Human blood and Milk of magnesia

(4) Antacid, Milk, Window cleaner, Glucose solution

38. Consider the following salts:-

(i) NaF

(ii) $NaClO_4$

(iii) $NaHSO_4$

Which of them when dissolved in water would form a basic solution?

(1) (i) only (2) (ii) & (iii) only (3) (iii) only (4) (i) & (iii) only

39. The two atoms having the same number of neutrons are considered as isotones. The below given pairs are the exact example of isotones. Determine the number of protons, neutrons and electrons present in them:

(a) Potassium and Calcium (b) Tritium and Helium

(1)

Atom	p	e^-	n
K	19	19	20
Ca	20	20	20
H	1	1	1
He	2	2	2

(2)

Atom	p	e^-	n
K	19	19	20
Ca	20	20	20
H	1	1	2
He	2	2	2

(3)

Atom	p	e^-	n
K	20	19	19
Ca	40	20	40
H	1	1	0
He	2	2	2

(4)

Atom	p	e^-	n
K	19	20	19
Ca	20	20	20
H	2	1	2
He	2	2	1

40. Which of the following statements is correct?

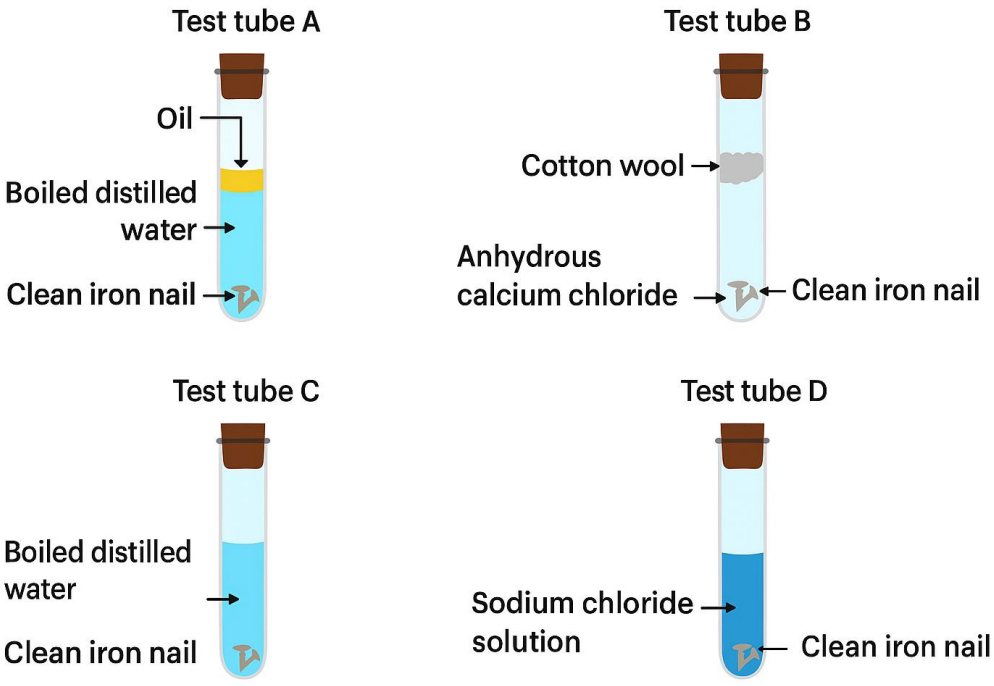
(1) The number of valence electron in sodium is 6.

(2) The atoms $_{18}X^{40}$ and $_{20}Y^{40}$ represent a pair of isotopes.

(3) Cl^{35} and Cl^{37} have same chemical properties.

(4) Electronic configuration of Calcium is 2, 8, 10.

41. Refer to the given figure and answer the following question:



Rusting of iron nail is taking place in which of the following test tubes?

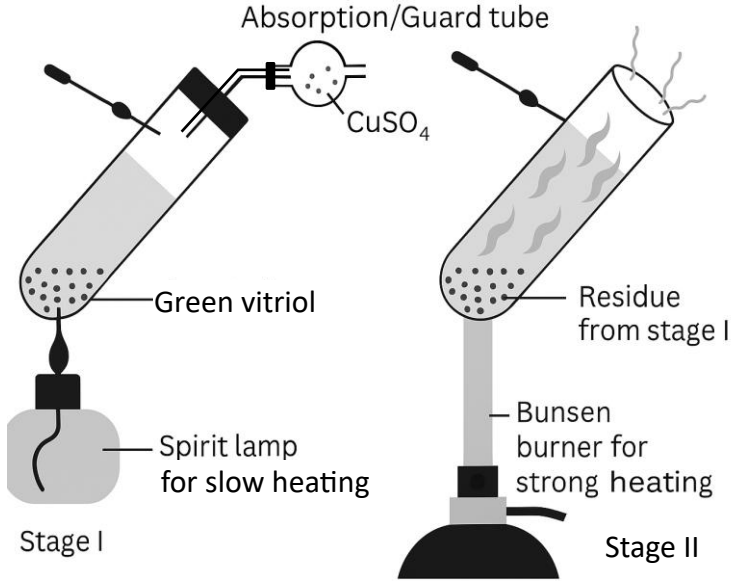
(1) A and C (2) B and C (3) A and D (4) C and D

42. What will be the value of a, b, c and d respectively in the following equation?

a $C_2H_4O_2$ + b Fe → c $Fe(C_2H_3O_2)_3$ + d H_2

(1) 6, 3, 3, 2 (2) 6, 2, 3, 1 (3) 6, 2, 2, 2 (4) 6, 2, 2, 3

43. Look at the experiment carried out by a student in two stages:



Stage - I :

(i) What will happen to anhydrous copper sulphate ?

(ii) What will be left behind in the test tube ?

Stage - II :

(i) Which gas/gases will be evolved ?

(ii) What will be the smell of the gas/gases ?

(iii) What will be the colour and formula of the residue left behind ?

Choose the correct option in Stage - I and Stage – II

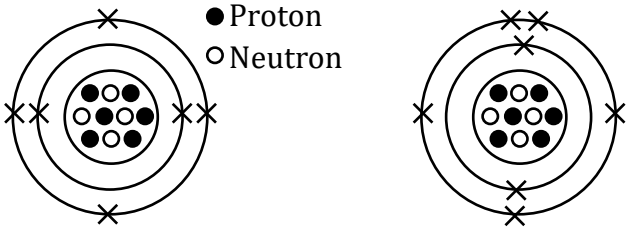
		Stage - I		**Stage - II**
(1)	(i)	No change.	(i)	Only SO_3 gas would be evolved.
	(ii)	$FeSO_4$ will change into green.	(ii)	No change in smell.
			(iii)	Colourless residue, $FeSO_4$
(2)	(i)	White anhydrous $CuSO_4$ will change to blue hydrated $CuSO_4.5H_2O$	(i)	A mixture of SO_2 and SO_3 gases will be evolved.
	(ii)	Anhydrous $FeSO_4$ will be left behind in the test tube.	(ii)	Smell like that of burning matchstick.
			(iii)	Reddish brown residue will be left behind, Fe_2O_3

(3)	(i)	No change.	(i)	SO_2 gas would be evolved.
	(ii)	$FeSO_4$ will change into black.	(ii)	Pungent smell.
			(iii)	Reddish yellow will be left, FeO.
(4)	(i)	Change into black copper oxide.	(i)	No change.
	(ii)	$FeSO_4$ will change into white.	(ii)	No change in smell.
			(iii)	White residue, PbO.

44. A compound 'C' of sodium is of white colour. It is a main constituent of baking powder, when heated it gives a gaseous compound 'D'. When gas 'D' is dissolved in water, it forms a weak acid 'G'. Which among the following is compound 'G' ?

(1) CH_3COOH (2) H_2PO_4 (3) H_2CO_3 (4) CH_3COONa

45. The given diagrams show the atomic structures of elements X and Y.



What name is given to the pair of elements X and Y?

(1) Isotopes (2) Isobars (3) Isotones (4) Isoelectronic

46. $^{14}_{6}R$ and $^{15}_{7}R$ are the symbol of two elements. They are

(1) Isotopes (2) Isotones (3) Isobars (4) Isoelectronic

47. Analyse the two statements given below and select the correct option.

Statement I: When the free ends of a tester are dipped in solutions containing dilute HCl and CH_3COOH turnwise, the bulb lights up more brightly in the first case.

Statement II: HCl is a stronger acid than CH_3COOH.

(1) Both Assertion (A) and Reason (R) are the true and Reason (R) is a correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are the true but Reason (R) is not a correct explanation of Assertion (A).

(3) Assertion (A) is true and Reason (R) is false.

(4) Assertion (A) is false and Reason (R) is true.

48. Study the following statements about dilute sulphuric acid.

(i) A white precipitate is formed when aqueous barium chloride is added.

(ii) The solution turns anhydrous copper(II) sulphate from white to blue.

(iii) Addition of a Universal Indicator shows that the solution has a pH value of less than 7.0.

(iv) The solution reacts with copper(II) oxide, forming a blue solution.

Which of the following statements confirm the acidic nature of the dilute sulphuric acid ?

(1) (i) and (ii) only (2) (i) and (iii) only

(3) (i), (iii) and (iv) only (4) (iii) and (iv) only

49. Study the following statements

(i) Calcination and roasting of ores are the steps of metallurgy.

(ii) Calcination and roasting reduce the ore to metal.

(iii) Metals at the top of the series are extracted by electrolytic reduction.

(iv) Metals at the top of the series are easily reduced.

Which of the following statements are correct ?

(1) (i) & (iii) only (2) (ii) & (iv) only (3) (ii) & (iii) only (4) All are correct

50. What will be the decreasing reactivity order when metals react with dilute mineral acid?

(1) Na > Zn > Mg > Fe > Cu (2) Na > Mg > Fe > Zn > Cu

(3) Na > Mg > Zn > Fe > Cu (4) Zn > Na > Mg > Cu > Fe

51. You are given the atom of an element $_{17}X^{35}$. What will be the chemical formula of the compound formed when X reacts with:

(a) Hydrogen

(b) Carbon respectively

(1) X_2H, XC_2 (2) HX_2, CX_2 (3) HX, CX_4 (4) HX, CX

52. An element X has atomic number 9 and mass number 19. Its anion is represented by

(1) X^{2-} (2) X^- (3) X^{2+} (4) X^+

53. The following reaction is used for the preparation of oxygen gas in the laboratory.

$2KClO_3(s) \xrightarrow{Heat} 2KCl(s) + 3O_2(g)$

Which of the following statement about the reaction is correct ?

(1) It is a decomposition reaction and endothermic in nature.

(2) It is a combination reaction.

(3) It is a decomposition and accompanied by release of heat.

(4) It is a photochemical decomposition reaction and exothermic in nature.

54. Which of the following compound imparts red colour to the flame and decomposes on heating to give oxygen and a brown gas ?

(1) $CaCO_3$ (2) $Mg(NO_3)_2$ (3) $MgCO_3$ (4) $Ca(NO_3)_2$

55. If 12g of carbon reacts with 4g of Hydrogen, then find the amount of Hydrogen required to react with 8g of carbon.

(1) 9 kg (2) 9 g (3) 2.66 kg (4) 2.66 g

56. A metal M forms a compound M_3PO_4. What will be the formula of its metal bicarbonate?

(1) $M(HCO_3)_4$ (2) $M(HCO_3)_2$ (3) $MHCO_3$ (4) $M(HCO_3)_3$

57. How many total protons are found in one molecule of retinol ($C_{20}H_{30}O$) ?

(1) 51 (2) 151 (3) 600 (4) 158

58. When the solution of a base is diluted what will be the change in pH of the solution?

(1) pH of the solution remains the same

(2) pH of the solution will increase

(3) pH of the solution will decrease

(4) pH of the solution climbs to 7

59. Acetic acid was added to a solid X kept in a test tube. A colourless and odourless gas Y was evolved.

The gas was passed through lime water which turned milky. It was concluded that

(1) Solid X is sodium hydroxide and the gas Y is CO_2.

(2) Solid X is sodium carbonate and the gas Y is CO_2.

(3) Solid X is sodium acetate and the gas Y is CO_2.

(4) Solid X is sodium hydrogen carbonate and the gas Y is SO_2.

60. The compound which aqueous solution will have the lowest pH is

(1) $NaHCO_3$ (2) NH_4Cl (3) Na_2CO_3 (4) NaCl

61. Assertion : Magnesium is extracted from its ore by electrolytic reduction.

Reason : All metals are extracted by electrolytic reduction.

(1) Both assertion and reason are correct and reason is the correct explanation of the assertion

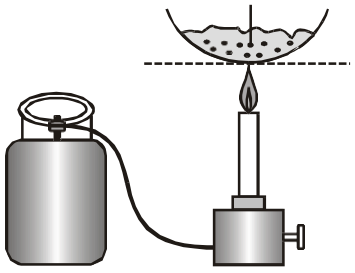
(2) Both assertion and reason are correct , but reason is not the correct explanation of the assertion.

(3) Assertion is correct, but reason is incorrect.

(4) Assertion is incorrect, but reason is correct.

62. Observe the mixture given below.

Powdered iron and copper (II) oxide



On heating the mixture strongly taken in a crucible,

(1) Formation of iron (II) oxide and copper takes place.

(2) Formation of copper (III) oxide and iron (II) oxide takes place.

(3) Formation of iron (III) oxide and copper (II) sulphate takes place.

(4) No reaction takes place.

63. The postulates of Bohr's atomic model are given below. Arrange them in the correct sequence.

(a) As long as the electron revolves in a particular orbit, the electron does not lose its energy. Therefore, these orbits are called stationary orbits and the electrons are said to be in stationary energy states.

(b) Electron revolve round the nucleus in specified circular path called orbits or shells.

(c) The energy associated with a certain energy level increase with the increase of its distance from the nucleus.

(d) An electron jumps from a lower energy level to a higher energy level by absorbing energy. But when it jumps from a higher to lower energy level, energy is emitted in the form of electromagnetic radiation.

(e) Each orbit or shell is associated with a definite amount of energy. Hence these are also called energy levels and are designated as K, L, M, N respectively.

(1) a c d e b (2) b c e a d (3) b e c a d (4) b a d c e

64. Which among the following are isobars?

(1) ${}_{b}X^{a}$, ${}_{b}Y^{a+1}$ (2) ${}_{b}X^{a}$, ${}_{c}Y^{b}$ (3) ${}_{b}X^{a}$, ${}_{b+1}Y^{a}$ (4) ${}_{b}X^{a}$, ${}_{b-1}Y^{a-1}$

65. In a proper equation, which are the only numbers that can be altered ?

(1) Molar masses (2) Superscripts (3) Coefficients (4) Subscripts

66. Match chemical reactions given in the List-I with the type of chemical reactions given in List-II and select the correction answer using the options given below:

	Column-I		**Column-II**
A	Formation of H_2O from H_2 and O_2	I	Decomposition
B	Heating of calcium carbonate	II	Single displacement
C	Reaction of aqueous $Pb(NO_3)_2$ solution with aqueous KI solution	III	Combination
D	Strip of iron placed in copper sulphate solution	IV	Double displacement

(1) A-III, B-I; C-IV, D-II (2) A-III, B-I, C-II, D-IV

(3) A-I, B-II, C-III D-IV (3) A-II, B-I, C-III, D-IV

67. There are two elements X and Y.

X : atomic number = 8, Mass number = 16

Y : atomic number = 8, Mass number = 17

Which of the following is correct about these two atoms?

(1) Electronic configuration of X is 2, 8, 6 and while that of Y is 2, 8, 7.

(2) Both X and Y are isobars.

(3) X has 2 shells while Y has 3 shells.

(4) Both X and Y have same chemical properties.

68. Which of the following is not a reduction process?

(1) $Fe^{2+} \longrightarrow Fe^{3+} + e^-$ (2) $Fe^{2+} + 2e^- \longrightarrow Fe$

(3) $Fe^{3+} + e^- \longrightarrow Fe^{2+}$ (4) $O_2 + 4e^- \longrightarrow 2O^{2-}$

69. Silicon is a chemical element discovered by Berzelius and it is present in sand as silicon dioxide. We know that the silicon atom contains 14 times more protons than hydrogen and that the number of electrons in its shell is equal to the number of neutrons. Which of the following statements is true about the particles in the atomic nucleus of silicon?

(1) it has 28 nucleons (2) it has 28 neutrons

(3) it has 28 electrons (4) it has 10 neutrons

70. The chemical formula of a substance shows:

(1) The actual number of atoms of each element in one molecule of a substance.

(2) The elements that make up one molecule of the substance and the fractional whole number ratio between the atoms.

(3) The number of molecules in a sample of the substance.

(4) The molecular mass of the substance

71. Which of the following contains both Mg & Ca ?

(1) Magnesite (2) Calamine (3) Carnallite (4) Dolomite

72. Which of the following is/are correct?

(A) On heating, the pH of water increases.

(B) Nature of solution is acidic when ammonium chloride is dissolved in water.

(C) Turmeric changes from yellow to brown in acidic solution.

(D) Ferrous sulphide and dil. H_2SO_4react and gives rotten egg smell gas.

(1) A only (2) A and B (3) B and D (4) A, B, C and D

73. The electronic configuration of a dipositive metal ion M^{2+} is 2, 8, 8, 4 and its atomic weight is 58 a.m.u. The number of neutrons in its nucleus would be-

(1) 30 (2) 32 (3) 34 (4) 42

74. Which of the following can undergo chemical reaction ?

(1) $MgSO_4 + Fe$ (2) $ZnSO_4 + Fe$ (3) $MgSO_4 + Pb$ (4) $CuSO_4 + Fe$

75. A metal object corrodes when it is exposed to the open air and humid weather. In iron, this change is called rusting in which flaky brown compound is formed. Which condition can cause severe rusting fast?

(1) Submerged in boiled water with oiled layer.

(2) Submerged in salt water.

(3) Exposed in vacuum with moisture.

(4) Exposed in air with calcium chloride.

76. The elements P, Q, R having given number of electrons in their respective outermost shells. Element P has 5 electrons in its L shell, element Q has 5 electrons in its M shell while element R has 7 electrons in its L shell. Which of the following statements is/are correct regarding elements P, Q and R ?

(I) P and Q have valency equal to 3.

(II) Atomic number of R and P are 7 and 9 respectively.

(III) R has least atomic size among three elements.

(IV) Element Q is stored in under Kerosene as it is highly reactive.

(V) Element R has 3 more neutrons than element P

(VI) Mass number of P is 28 and R is 18.

(1) I, III, V, VI (2) I, II, III, V (3) I, III, IV, V (4) I, III, V

77. A thin sheet of gold foil is bombarded with α-particles as in Rutherford's experiment. Which of the given descriptions most accurately represents the path of α-particles?

	Undeflected	**No. of particles deflected by a small angle**	**No. of particles appeared to rebound**
I	All	None	None
II	Most	Most	None
III	Most	Some	Very few
IV	A few	Very few	Some

(1) I (2) II (3) III (4) IV

78. Read the following statements & select the option that correctly fill the blanks.

(i) The reaction in which one of the reactant carries the reduction of the product is known as_____

(ii) Heating ore in the presence of excess of air is called_____

(iii) Reduction of metal oxides by Aluminium is known as_____

(iv) Heating ore in the absence of air is known as_____

	(i)	(ii)	(iii)	(iv)
(1)	Heating	Roasting	Thermite process	Calcination
(2)	Smelting	Calcination	Thermite process	Roasting
(3)	Self reduction	Roasting	Bayer's process	Calcination
(4)	Auto reduction	Roasting	Aluminothermy	Calcination

79. Which of the following is not redox reaction?

(1) $2Mg + O_2 \rightarrow 2MgO$ (2) $CaCO_3 \rightarrow CaO + CO_2$

(3) $H_2 + Cl_2 \rightarrow 2HCl$ (4) $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$

80. A compound 'X' on heating produces a colourless gas 'Y' and metal oxide 'Z'. The metal oxide is yellow when hot and show white colour when it is cooled. Identify X, Y & Z:

(1) X = $ZnCO_3$, Y = CO_2, Z = ZnO (2) X = $CaCO_3$, Y = CO_2, Z = CaO

(3) X = $AgNO_3$, Y = NO_2, Z = AgO (4) X = $CuSO_4$, Y = SO_2, Z = CuO

81. In the word "NICe", identify the symbols of the chemical elements.

(1) nitrogen, iodine, caesium (2) nickel, caesium

(3) nitrogen, caesium (4) nitrogen, iodine, cerium

82. In order to balance the chemical equation

$H_2O_2 + KMnO_4 + HCl \rightarrow MnCl_2 + KCl + H_2O + O_2$

The stoichiometric coefficients are

(1) 5,2,2,2,2,8,2 (2) 2,5,2,5,5,8,2 (3) 2,2,5,2,2,5,1 (4) 5,2,6,2,2,8,5

83. Description of a few atoms are given in the table.

Atom	P	Q	R	S
No. of protons	17	11	17	9
No. of neutrons	18	12	20	10
No. of electrons	17	10	17	10

Identify a cation, an anion and a pair of isotopes from the given table.

	Cation	Anion	Pair of isotopes
(1)	P	Q	P, Q
(2)	Q	S	P, R
(3)	R	P	R, S
(4)	Q	S	Q, S

84. A salt 'C' of sodium is of white colour. It is used for making baking powder, also used as an antacids and in fire extinguishers.

(1) CH_3COONa (2) $Na3PO_4$ (3) $NaHCO_3$ (4) NaCl

85. Which of the following ores are calcinated during extraction? Select the correct option.

(a) Copper pyrites (b) Cinnabar (c) Halite

(1) a, b & c (2) b & c (3) Only a (4) None of these

86. Consider the following four equations

I. $HS^- + H_2O \rightleftharpoons S^{2-} + H_3O^+$

II. $HS^- + H_2O \rightleftharpoons H_2S + OH^-$

III. $NO_3^- + H_2O \rightleftharpoons HNO_3 + OH^-$

IV. $HCO_3^- + H_2O \rightleftharpoons H_2CO_3 + OH^-$

Water has an acid character in equation

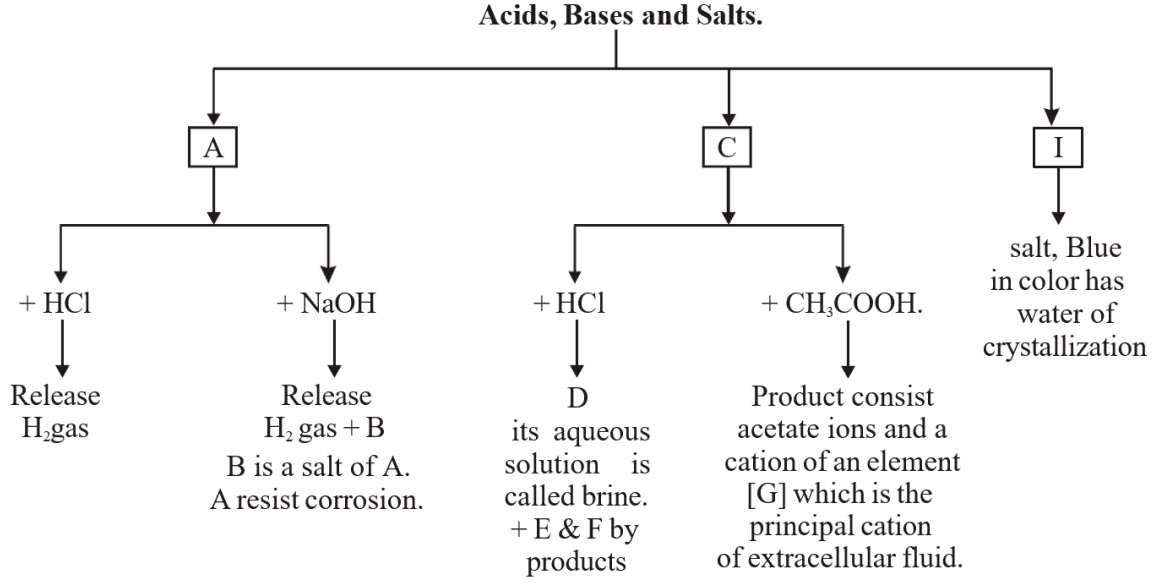
(1) I, II, III, IV (2) I, II, III (3) II, III, IV (4) I, II

87. P is produced by the action of chlorine on dry slaked lime. Q is a non-corrosive base and used for faster cooking. On heating R at 373 K, it becomes calcium sulphate hemihydrate. The P, Q and R respectively are

(1) $CaOCl_2$, $NaHCO_3$, Gypsum (2) Gypsum, Na_2CO_3, $CaOCl_2$

(3) $Ca(OH)_2$, $NaHCO_3$, Plaster of paris (4) $CaOCl_2$, Na_2CO_3, NH_4Cl

88.



(1)	A-Zn,	F-CO_2	B-Na_2ZnO_2,	C-NaOH.
(2)	A-Al,	E-H_2O,	B-$NaAlO_2$,	G - K
(3)	A-Sn,	C-Na_2CO_3,	I-$CuSO_4.5H_2O$,	D-NaOH
(4)	A-Sn,	C-Na_2CO_3,	D-NaCl,	G-Na

89. When a sample of copper containing iron as impurity is purified by electrolysis, the appropriate electrodes are :

	Cathode	**Anode**
(1)	Pure iron	Pure copper
(2)	Impure sample	Pure copper
(3)	Impure iron	Impure sample
(4)	Pure copper	Impure sample

90. When a metal carbonate 'X' is heated strongly, it decomposes into metal oxide 'Y' and a gas 'Z'. The metal oxide 'Y' is a major constituent of cement. Identify X, Y & Z respectively

(1) $CuCO_3$, CuO, CO_2 (2) $ZnCO_3$, ZnO, CO_2

(3) $CaCO_3$, CaO, CO_2 (4) $MgCO_3$, MgO, CO_2

91. There are three elements I, II and III.

I : It is the largest among three elements with valency two.

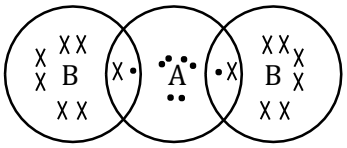
II : It has 1 more neutron than that of I.

III : It has 2 shells and 4 valence electrons.

By using the given hints. Identify elements, I, II and III respectively.

(1) B, N, C (2) O, F, Ne (3) Be, B, C (4) Mg, B, C

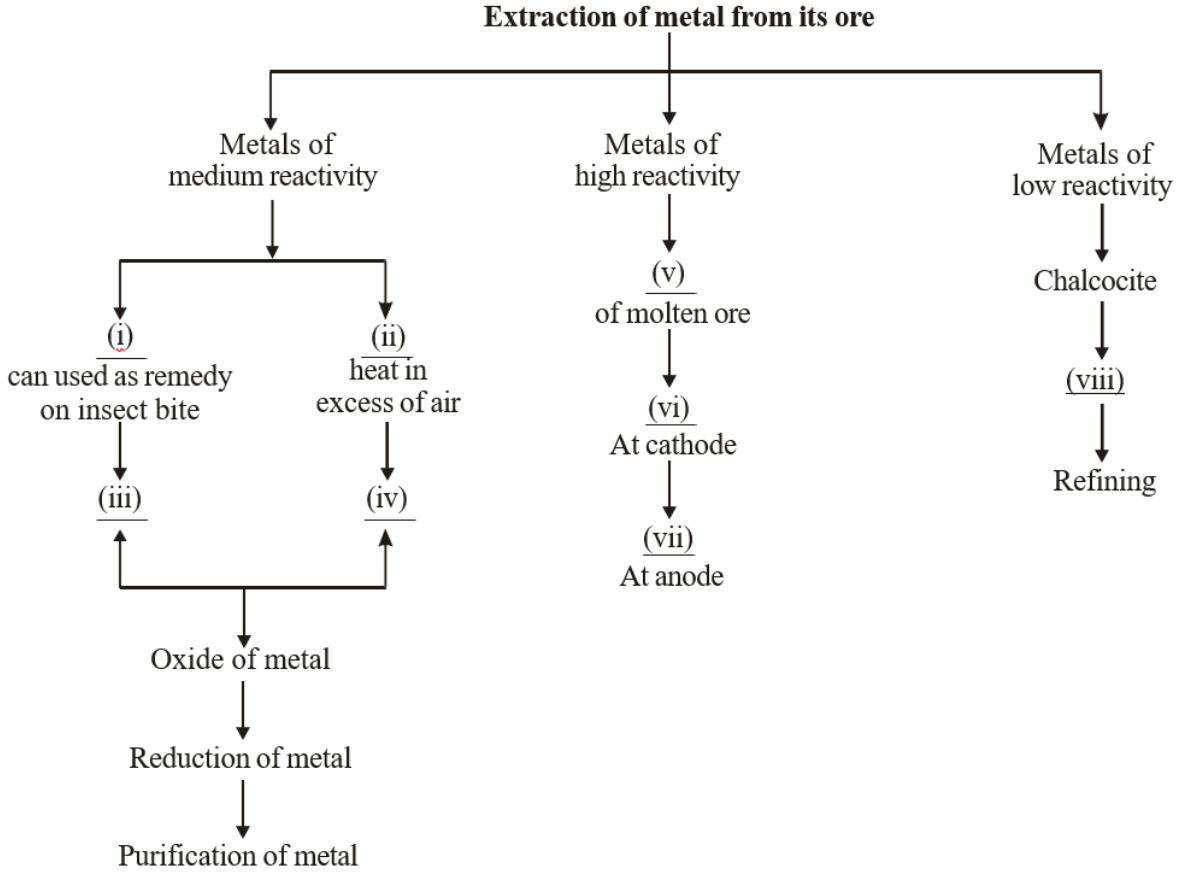
92. A compound, AB_2 has the following arrangement of electrons



The elements A and B are respectively.

(1) F, O (2) S, O (3) Ar, O (4) Xe, F

93. Metals are extracted from their ores by using different methods depending on their position in the activity series. Fill in the blanks in the given flow chart by choosing an appropriate option.



(1)	(i)-$CaCO_3$,	(iv) Calcination,	(viii) Roasting,	(vii) Positive ion
(2)	(ii)-Zns,	(v) Electrolytic refining,	(iii) Roasting	(vi) Positive ion
(3)	(i)-$ZnCO_3$,	(iv) Roasting,	(viii) Roasting	(vi) Positive ion
(4)	(ii)-Cu2S,	(iii) Calcination,	(v) Electrolytic refining	(vii) Negative ion

94. A student mixed two solutions X & Y. He recorded the following observations.

(i) A white precipitate is formed.

(ii) One of the product formed can undergo sublimation.

The solution X & Y are respectively

(1) $AgNO_3$(aq) and NaI aq) (2) $AlCl_3$(aq) and NH4OH(aq)

(3) FeS(s) and H_2SO_4(aq) (4) $CuSO_4$(aq) and NaOH(aq)

95. Oxide 'M' is a colourless gas. It reacts with oxygen to form a reddish-brown gas, Oxide 'N'. Oxide 'N' is also formed by the decomposition of a compound 'X' along with the formation of a yellow residue 'Y'. Which of the following statement is correct regarding to above information?

(1) Oxide 'M' is PbO and oxide 'N' is PbO_2.

(2) In conversion of oxide 'M' to oxide 'N', oxide 'M' acts as oxidising agent.

(3) In conversion of oxide 'M' to oxide 'N', oxide 'M' act as reducing agent.

(4) Oxide 'M' is CO and oxide 'N' is CO_2.

96. Which of the following is the formula of the compound nickel bisulphate?

(1) $NiHSO_4$ (2) Ni_2HSO_4 (3) Ni_2SO_4 (4) $Ni(HSO_4)_2$

97. Many elements have non-integer atomic masses because

(1) they have isobars.

(2) their isotopes have non-integral masses.

(3) their isotopes have different masses.

(4) their constituents, electrons, protons and neutrons combine to give fractional masses.

98. An atom has 2, 8, 8 and 2 electrons in its K, L, M and N orbits. What will be the symbol of element?

(1) Ca (2) Cu (3) C (4) Ar

99. When a gold sheet is bombarded by a beam of α - particles, only a few of them get deflected whereas most passes without any deviation. This is because

(1) the force of attraction exerted on the α - particle by the oppositely charged electrons is not sufficient.

(2) a nucleus has a much smaller volume than that of atom.

(3) the force of repulsion acting on the fast moving α - particles is very small.

(4) the neutrons in the nucleus do not have any effect on the α - particles.

100. A salt 'x' is used to treat hardness of water and also used for the preparation of Borax. Which of the following is correct chemical name for the salt 'x'?

(1) Sodium bicarbonate (2) Sodium carbonate decahydrate

(3) Sodium carbonate hemihydrate (4) Copper carbonate

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**2**	**3**	**3**	**3**	**2**	**1**	**3**	**3**	**2**	**3**	**3**	**3**	**2**	**3**	**4**
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	**3**	**4**	**3**	**4**	**3**	**1**	**2**	**2**	**3**	**1**	**3**	**2**	**1**	**3**	**4**
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	**1**	**2**	**1**	**1**	**3**	**1**	**3**	**1**	**2**	**3**	**4**	**4**	**2**	**3**	**2**
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	**2**	**1**	**4**	**1**	**3**	**3**	**2**	**1**	**4**	**4**	**3**	**4**	**3**	**2**	**2**
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	**3**	**1**	**3**	**3**	**3**	**1**	**4**	**1**	**1**	**1**	**4**	**3**	**3**	**4**	**2**
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	**4**	**3**	**4**	**2**	**1**	**4**	**4**	**2**	**3**	**4**	**3**	**1**	**4**	**4**	**3**
Question	91	92	93	94	95	96	97	98	99	100					
Answer	3	**4**	**3**	**2**	**3**	**4**	**3**	**1**	**2**	**2**					